Nagaraju Karan

Machine Learning Engineer

In the dynamic field of technology, I am an experienced professional with 6+ years of focus on Artificial Intelligence (AI) and Machine Learning (ML). This includes using natural language processing (NLP) and Computer vision enhance user interactions and applying machine learning algorithms for predictive analytics. My specialization in ML engineering positions me at the cutting edge of technological innovation. Additionally, I have also worked on iOS app development, leveraging my skills to create robust and user-friendly mobile applications.

Work History

2025-04 - Current

Senior Software Engineer

ProcureX Technologies, Hyderabad

- Designed and developed **scalable iOS applications** for **procurement and construction project management** using **Swift, SwiftUI, Combine**, and **MVVM architecture**.
- Built and maintained **Inventory, PMS, and Procurement modules**, ensuring timely delivery and high-quality releases.
- Leveraged **Combine** for reactive data handling, state management, and real-time UI updates to improve app responsiveness.
- Improved app performance through **UI optimization, memory management**, and crash reduction.
- Implemented **unit and UI testing** using **XCTest**, increasing release stability.

2024-09 - 2025-03

Senior Software Engineer

NeuroGaint Systems, Hyderabad, India

- Led the development of an AI-driven **Preventative Mental Health** iOS app, integrating sentiment analysis and emotion recognition for early detection of mental health issues.
- ● Designed conversational AI systems for personalized mental health support and actionable insights.

Contact

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Skills

Python, PyTorch (Deep Learning and Artificial Neural Networks), TensorFlow, NumPy and Pandas.

Supervised and unsupervised learning Algorithms, GenAI, Linear Algebra, Probability & Statistics.

Natural Language Processing (Text Classification, Sentiment Analysis, Machine Translation), Retrieval-Augmented Generation(RAG).

Large Language Models (LLM), Transformers Architecture, BERT, GPT Models in Natural Language Processing and Computer vision.

Model Fine-Tuning, Transfer Learning, Object detection, image segmentation and instance segmentation using Convolutional Neural

- ● Built scalable NLP pipelines and fine-tuned models for analyzing sensitive mental health data while ensuring privacy compliance.
- ● Collaborated with psychologists to align AI solutions with real- world mental health requirements.
- ● Enhanced the app with voice-based assessments and interactive features, improving accessibility and user engagement.

2022-07 - 2024-05

Senior Software Engineer

IGURU PORTAL SERRVICES, Hyderabad, india

- Designed and implemented advanced NLP models to enhance text analysis and processing capabilities, improving accuracy and efficiency by 25%.
- Developed custom transformer models and successfully deployed them into Hugging Face Spaces, facilitating real-time NLP solutions.
- Created robust text classification systems to distinguish between AI-generated and human-generated content, achieving an accuracy rate of over 90%.
- Utilized machine learning algorithms and NLP techniques to categorize text data, leading to improved data management and retrieval processes.
- Developed and optimized machine translation models to support multilingual content, enhancing global communication and user experience.
- Leveraged deep learning techniques to improve the quality and accuracy of translations across multiple languages.

2020-07 - 2022-02

Senior Software Engineer

Promantra Synergy Solutions Limited, Hyderabad, india

- Applied Agile software development methodology to elevate final products and streamline development efforts.
- Translated basic design plans into full iOS app frameworks, delegating project components to appropriate team members.
- Designed quality assurance testing for junior developer output, maintaining adherence to business goals and boosting productivity.
- Researched and selected APIs for integration into development projects.

Network(CNN).

AWS, Docker, FastAPI and Hugging Face.

Xcode, IOS, Swift ans SwiftUI. Payment gateway integration.

Languages

English

Hindi

Telugu

2018-07 - 2020-05

IOS Application Developer

Vitruvian 3D Solutions Pvt Ltd, Hyderabad, india

- Designed user experience frameworks applicable to fit both iPad and iPhone screens.
- Worked with quality assurance team to confirm app compatibility with analogous solutions on other platforms.
- Applied Agile software development methodology to elevate final products and streamline development efforts.
- Included visible but unobtrusive ad placement opportunities within apps, increasing revenue potential without disrupting usage.

Education

2016-01

Bachelor of Engineering: Computer Science And Programming

Matrusri Engineering College - Hyderabad, TG

2012-01

MPC: Maths, Physics And Chemistry

Intermediate Junior College

2010-01

Secondary School Certificate

ZPHS High School - Kondapur

Projects

OCR-Based Intelligent Invoice Extraction using Donut Transformer

- Designed and built an **end-to-end OCR and document understanding system** using **Donut (Document Understanding Transformer)** to extract structured data from **diverse invoice formats** across vendors and industries.
- Developed a **layout-aware vision-language model** that directly converts invoice images and PDFs into **structured JSON outputs** without traditional OCR, ensuring robustness on noisy and unstructured documents.
- Fine-tuned the model on **custom multi-layout invoice datasets** (GST, retail, logistics, service invoices), enabling strong generalization across unseen formats.

- Implemented **schema-driven extraction and preprocessing pipelines** to accurately capture key fields such as vendor details, GSTIN, line items, taxes, and totals.
- Integrated **human-in-the-loop validation** and deployed the solution as a **scalable API** with ERP/accounting system integration, achieving **95%+ field-level accuracy** and significantly reducing manual data entry.

Deep Learning Model for CIELAB Color Prediction Based on Human Features:

- Developed a deep learning model leveraging convolutional neural networks (CNN) to predict CIELAB color values of a person based on a group of their images.
- Incorporated **categorical input features** such as age, gender, and ethnicity to enhance prediction accuracy.
- Processed diverse datasets to ensure robust and inclusive predictions across various demographic groups.
- Implemented **data preprocessing pipelines** for handling image noise and ensuring high-quality input for training.
- **Architecture**: Convolutional Neural Network (CNN) designed for image-based feature extraction and regression tasks.
- **Evaluation**: Achieved an impressive **95% accuracy** in predicting CIELAB values after rigorous training and validation.
- **Optimization**: Conducted hyperparameter tuning, regularization, and augmentation to improve model performance and generalization.
- **Impact**: The model provides a foundation for applications in personalized color recommendations, digital fashion, and cosmetic industries.

Languages

6,3,B1,5,C1